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Electronics Lab
इलेक्ट्रॉनिक्स लैब

Experiment Number 6
प्रयोग संख्या 6

To study the 555-timer as Mono-stable and Astable Multivibrator.

555-टाइमर का मोनो-स्टेबल और एस्टेबल मल्टीवाइब्रेटर के रूप में अध्ययन करना।

MAULANA AZAD NATIONAL INSTITUTEOF TECHNOLOGY, BHOPAL

ELECTRONICS LAB

EXPERIMENT NO.6

Aim : To study 555 timer as

1. Astable Multivibrator
2. Monostable Multivibrator
3. Bistable Multivibrator
4. Voltage to Time convertor / Voltage to Frequency Convertor. (Application of IC 555)

Instrument required:

- DC Regulated Power Supply of +5V (Vcc) available on sockets.
- IC 555 placed inside the cabinet & connections brought out at sockets.
- Various resistances & capacitors are placed behind the panel & connections are brought out on the sockets.
- Two Push-to-ON switches are provided on the front panel, active low and active high for trigger input.
- One potentiometer VR1 is also mounted on the front panel to perform V to for V to T experiments.

Theory:

With the monolithic integrated circuit 555 we can get accurate timing ranges of microseconds to hours, independent of supply voltage variations. This versatile device has a large number of interesting practical applications.
Basically, the 555 timer is a highly stable integrated circuit capable of functioning as an accurate time-delay generator and as a free running multivibrator, when used as an oscillator the frequency and duty cycle are accurately controlled by only two external resistors and a capacitor. The circuit may trigger a 1 reset on falling waveforms. Its prominent features are

1. Timing from micro seconds through hours.
2. Monostable and stable operation.
3. Adjustable duty cycle.
4. Ability to operate from a wide range of supply voltages.
5. Output compatible with CMOS, DTL & TTL.
6. High current output can sink and source 200mA.
7. Trigger and reset inputs are logic compatible.
8. Output can be operated normal on and normal off.
9. High temperature stability

Procedure:

Astable multivibrator :-

1. Connect the circuit as shown in fig(1).
2. Connect CRO lead across pin no. 3 of 555 & ground point as shown in the circuit diagram.
3. Switch ON the instrument using the ON/OFF toggle switch provided on the front panel and also switch ON the CRO.
4. Observe the square wave output on CRO.
5. Calculate the frequency of output signal using formula

$$F = \frac{1.44}{(R_A + 2R_B)C_1}$$

6. Calculate duty cycle using the formula

$$D = \frac{R_B}{R_A + 2R_B}$$

- We can also perform this experiment on LEDs. For this connect the LED across output instead of CRO & change the value of C_1 to 1μF. The blinking LED will show the astable operation there is no stable state & the blinking will continue.

Circuit:

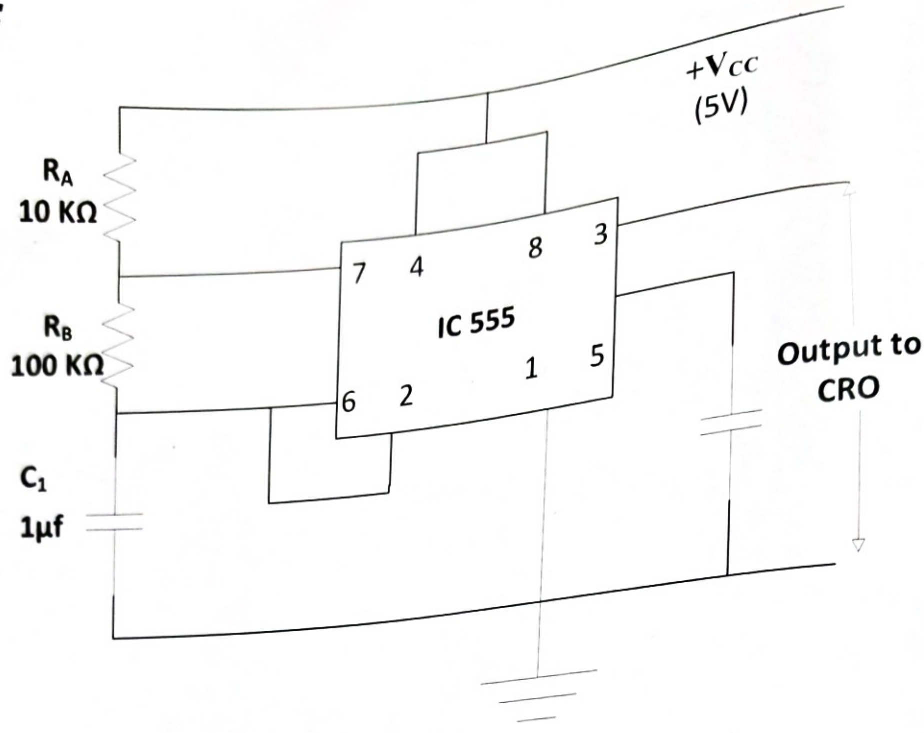

FIG 1. ASTABLE MULTIVIBRATOR

Monostable multivibrator: -

1. Connect the circuit as shown in fig(2).
2. Connect CRO lead across pin no. 3 of 555 & ground point as shown in the circuit diagram.
3. Connect Audio Frequency Signal Generator at the trigger input pin (Pin no 2 of 555) Set the Signal Generator output at square wave of 2V peak to peak amplitude, 1kHz frequency.
4. Switch ON the instrument using ON/OFF oggle switch provided on the front panel and also switch ON the CRO.
5. Observe the square wave output on CRO & calculate the Pulse Width time duration of high O/P of output using formula:

$$W=1.1R_AC_1$$

6. Now change the Capacitor C_1=1μF & observe the effect.
7. Connect the CRO across pin no. 6 of IC 555 & observe the output wave shape. It should be a saw tooth wave.

- We can also perform this experiment on LED. For this connect the LED across output instead of CRO & change the value of C_1 to 1μF. Also disconnect the signal generator from trigger input (pi no 2 of IC 555). Apply a trigger input high at pin no. 2 through push to on switch & observe the effect of trigger input on the glowing LED.

Circuit :

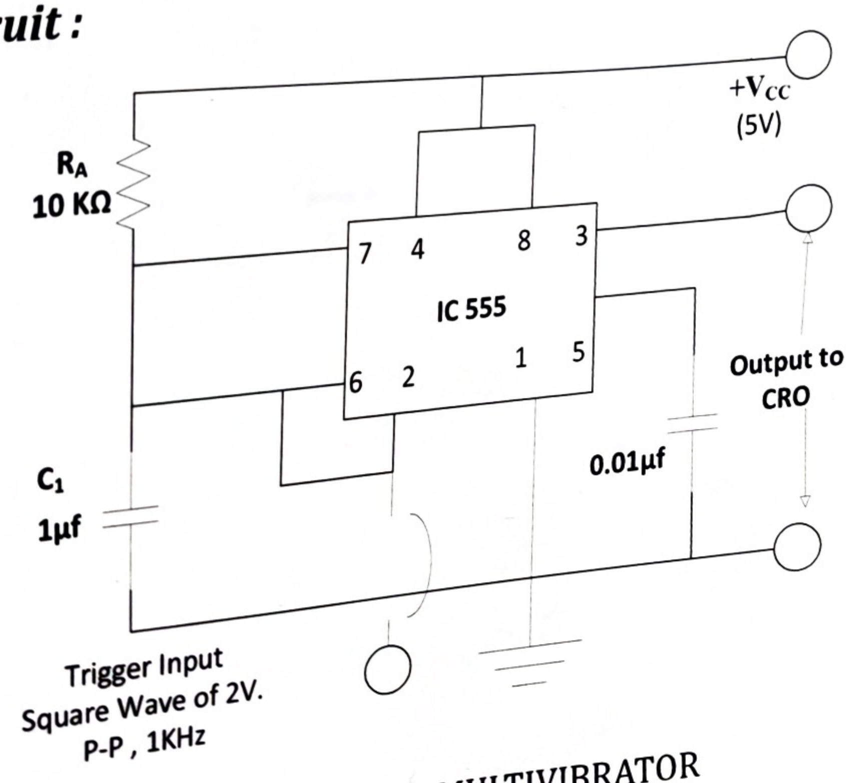

FIG 2. MONOSTABLE MULTIVIBRATOR

Bistable multivibrator:-

555 timer can also function as a bistable multivibrator. This multivibrator offers the advantage that it operates from many different supply voltages, uses little power and requires no external component other than bypass capacitors in noisy environments. It also provides a direct relay driving capability. As shown in circuit diagram a negative pulse applied to the trigger input terminal (pin no. 2) set the multivibrator and the output Q goes high. A positive going pulse applied to the threshold terminal will reset the multivibrator and drive the Q output low.

1. Connect the circuit as shown in fig. (3).

2. Switch ON the instrument using ON/OFF toggle switch provided on the front panel.

3. Apply a positive going pulse at pin no. 6 of IC, through push to on switch provided on front panel. It will reset the multivibrator and output Q geos low (LED off).

4. To set the multivibrator, apply a negative going pulse at pin no. 2. It will set the multivibrator and output Q goes high (LED glows).

Circuit :

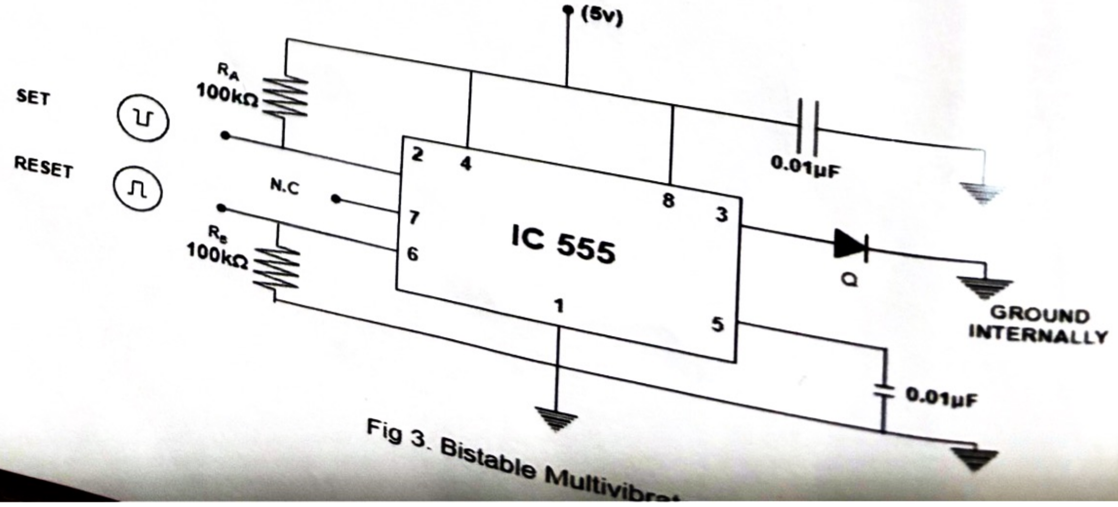

Fig 3. Bistable Multivibr

VOLTAGE TO TIME CONVERTOR / VOLTAGE TO FREQUENCY CONVERTOR:

1. Connect the circuit as shown in fig (4).

2. Connect external DC voltmeter between pin no. 5 & ground as shown in the Fig (4). Also connect CRO across pin no. 3 & ground point.

3. Switch ON the instrument using ON/OFF toggle switch provided on the front panel and also switch ON the CRO.

4. Observe the output wave form on CRO. Vary the voltage at pin no. 5 through potentiometer VR18 note down the corresponding change in time period & frequency of the output wave shape.

5. Calculate the time period using the formula

$$T = C_1(R_A + R_B)\left[\frac{(V_{CC} - V_{EXT})}{2}(V_{CC} - V_{EXT})\right] + 0.6931C_1R_B$$

6. Calculate the frequency by using formula F=1/T

Circuit :

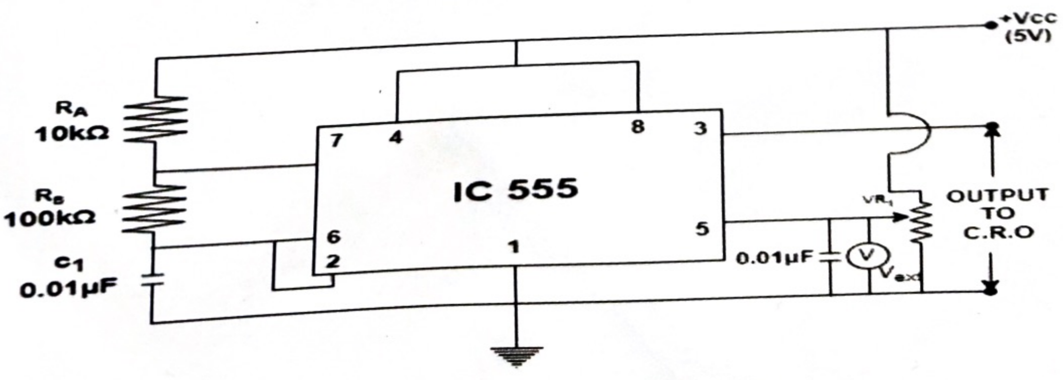

Fig 4. Voltage To Time Converter / Voltage To Frequency Converter

Standard Accessories:

1. Single point (4mm) Patch Cords for Interconnections. -10 Nos
2. Inter connectable (4mm) Patch Cords for Interconnections. -02 Nos
3. Instruction Manual (DOC 650). -01 Nos

Result & Conclusion:

Questions:

1. What is the purpose of the 555 timer in an astable multi vibrator mode?
2. What is the function of the 555 timer in monostable mod
3. What is the role of the 555 timer in bistable multivibrator mode?
4. How can a 555 timer be used as a voltage-to-time converter?
5. How does a 555 timer operate as a voltage-to-frequency converter?